

Based on detailed examination of the "energy.pdf" database listing Energy Science and Engineering professors at IIT Bombay, their research interests and publications, here is a refined, comprehensive superset of current and recent research topics they are working on:

Prof. Anish Modi focuses on thermal energy systems including solar thermal energy, power cycles, and polygeneration plants, refrigeration and heat pump systems, with research on advanced solar thermal utilization and energy sustainability.

Prof. Asish Kumar Sarangi's expertise lies in alternative fuels for internal combustion engines, advanced low-temperature combustion techniques, exhaust after-treatment catalysis, engine performance optimization, machine learning-based combustion phasing control in dual-fuel engines, and emission reduction strategies.

Prof. K. R. Balasubramaniam works on defect chemistry and bandgap engineering of transparent conductors and materials for photovoltaic and photoelectrochemical applications, with research on pulsed laser deposition of oxide thin films, X-ray scattering characterization of thin films, and oxygen exchange kinetics in complex oxides.

Prof. Chetan Singh Solanki's research includes silicon thin-film solar cells, photovoltaic module technologies, carbon nanotube synthesis and characterization, light absorption in multilayered solar absorbers, and optimization of catalyst loading for CNT growth.

Prof. D. Venkatramanan specializes in power electronics for energy storage and grid integration, including modeling and control of inverter-based resources, ultracapacitor stack design and performance, DC grid stability, dynamic phasor modeling, and solar photovoltaic grid-tied inverter architectures.

Prof. Dayadeep S. Monder's focus is first-principles catalysis, electrode and cell modeling for photoelectrochemistry, flow battery modeling, hydrogen evolution catalysis, CO₂ reduction mechanisms, and computational thermodynamics of energy materials.

Prof. Gurubalan Annadurai researches heating, ventilation, and air conditioning (HVAC) systems, energy recovery in membrane heat exchangers, transient modeling of desiccant beds, ultrasound-assisted performance improvement of membrane energy exchangers, and experimental approaches to fixed-bed regenerators.

Prof. J. K. Nayak works on solar thermal energy, passive solar architectural design, and solar domestic hot water systems including experimental evaluation and standard implementation.

Prof. Karthik Sasihithlu investigates light-matter interaction, metamaterials, phonon-assisted heat transfer across vacuum gaps, van der Waals force-assisted radiative transfer, thin film solar photovoltaics, and thermal radiation engineering.

Prof. Lalit Kumar specializes in rheology and multiphase flow, including oil wax deposition and shut-in/restart processes, thermal management of batteries and living spaces, flow optimization

using AI-assisted multi-objective algorithms, Rayleigh–Bénard convection in complex cavities, and thixotropic constitutive modeling.

Prof. Manaswita Bose focuses on granular materials flow dynamics, molecular dynamics simulations, Brazil nut effect, vibrated granular matter, and particulate flow modeling applied to coal gasification and combustion.

Prof. Manoj Neergat researches fuel cells including direct methanol and polymer electrolyte membrane fuel cells, flow batteries, electrochemical impedance spectroscopy, catalyst development for oxygen reduction reactions, and conducting polymer electrolyte membranes.

Prof. Prakash Chandra Ghosh's work centers on hydrogen production from renewable sources, hybrid fuel cells, CFD modeling of fuel cell flow fields, non-destructive testing methodologies of fuel cells, membrane-based humidifier development, and progress in high-temperature polymer electrolyte fuel cells.

Prof. Pratibha Sharma investigates solid-state hydrogen storage materials, metal hydride reactor simulation and heat transfer optimization, ammonia borane decomposition kinetics, porous carbon derivatives for hydrogen storage, and microstructural effects on hydrogen absorption in alloys.

Prof. Rajesh Gupta studies solar photovoltaic cell and module diagnostics, reliability, performance evaluation using electroluminescence and thermography techniques, crack impact on module temperature distribution, and floating photovoltaic system development.

Prof. Rangan Banerjee focuses on energy management and efficiency, demand side management, energy modeling, cogeneration, fuel cell and hydrogen storage systems, renewable energy policy and technology assessment, and integrated energy-economy-environment modeling.

Prof. Sagar Mitra's research spans lithium-ion and sodium-ion batteries, electrode and electrolyte materials, solid-state batteries, metal-sulphur batteries, commercial scale batteries, advanced battery electrodes with fluorine-rich electrolytes, and stabilization strategies for solid electrolytes.

Prof. Sandeep Kumar works on biomass gasification, hydrogen production via oxy-steam gasification, solid waste management through gasification, numerical simulation of co-gasification processes, and synthesis of H₂-enriched syngas from bio-waste.

Prof. Sankara Sarma V. Tatiparti researches hydrogen storage, supercapacitors, electrochemical nanostructured materials fabrication, scanning electrochemical microscopy, and polycrystalline nano-dendritic growth.

Prof. Santanu Bandyopadhyay's expertise includes process integration and pinch analysis for industrial energy conservation, carbon capture utilization and storage systems targeting optimization, heat-integrated water regeneration networks, and energy system modeling.

Prof. Shaibal K Sarkar focuses on perovskite solar cells, atomic layer deposition of photoabsorbers, enhancements in operational stability and encapsulation methods, effect of phase transitions on perovskite materials, and plasma-assisted coating processes.

Prof. Shireesh K. Kedare researches concentrating solar technologies with thermal energy storage, renewable energy technologies for decarbonization of Indian process industries, and energy transition pathways via systems modeling and participative development.

Prof. Sidda Vatam R. Prakash Reddy specializes in high-performance control of motor drives for electric vehicles, motor parameter identification, multilevel DC-DC converter topology for fast EV charging, and resistive superconducting fault current limiters integrated in DC hybrid circuits.

Prof. Surya Narayana Doolla's work centers on smart grid control and optimization, microgrid energy management using demand response, distributed energy resource management through multi-agent systems, and transient response improvement via virtual inertia.

Prof. V.S.S. Pavan Kumar Hari works on control of electrical machines, power electronics for wind energy conversion systems, magnetic gear design, railway power systems, micro-energy harvesting, and hybrid electric vehicle motor drives.

Prof. Zakir Hussain Rather focuses on grid integration of renewables, power system dynamics and control, wind and solar power plant control strategies, grid stability enhancement, and voltage and frequency regulation ancillary services.

Prof. Venkatasailanathan Ramadesigan researches battery performance and degradation modeling, fuel cell simulation, flow configuration effects on fuel cells, standalone renewable hybrid systems with physics-based models, and optimization of lithium metal batteries.

Prof. Vishnulal studies biodiesel synthesis from renewable feedstocks over catalysts including spent FCC, thermodynamic analysis of LNG regasification, and methanol synthesis using reactive distillation processes.

Prof. Suneet Singh's expertise includes nuclear reactor stability analysis, neutron diffusion modeling, flow instabilities in thermal systems, and non-intrusive measurement techniques for multiphase flows.

Prof. Zakir Hussain rather specializes in voltage control ancillary services in wind plants, wind farm control strategies to mitigate voltage dips and frequency excursions, wind turbine selection for farms, and grid stability enhancement through fault-ride-through retrofits.

This comprehensive and detailed summary encompasses the refined and specific current and recent research activities of the Energy Science and Engineering faculty at IIT Bombay. It reflects a broad and deep specialization ranging from solar energy systems, advanced batteries, fuel cells, thermal and HVAC systems, biomass utilization, policy and systems modeling, to power electronics and grid integration technologies.

1. <https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/attachments/126871353/a9e50ba5-416e-4666-b0a9-1f5f71e13c14/energy.pdf>